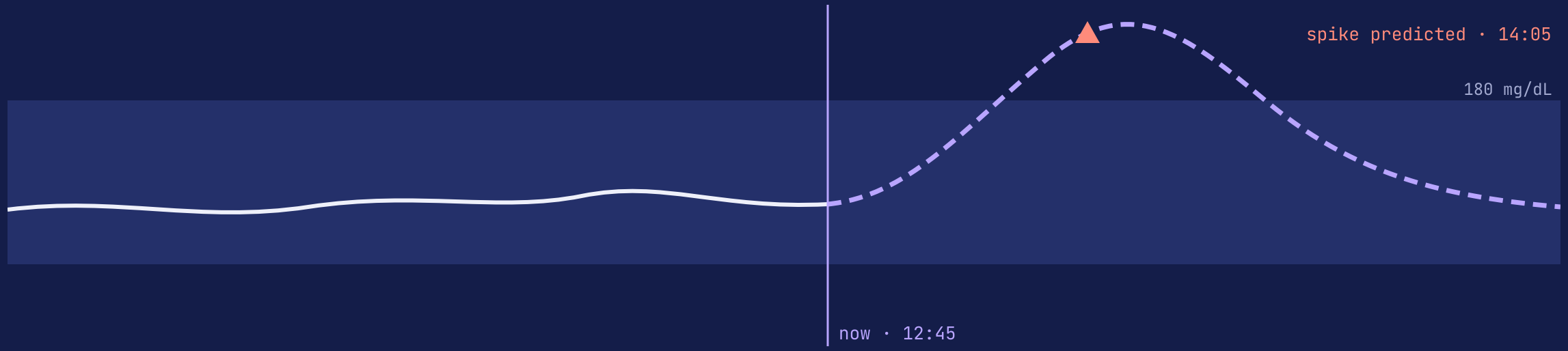
DIGITAL TWIN CHALLENGE 2026 · HAPPIEST HEALTH · SIMULATED DEMO

CHAYA छाया

The shadow that walks two hours ahead: a personal digital twin that sees a glucose spike **2 hours** before it happens.



THE PROBLEM

India's diabetes care runs on one number every three months.

101 M

Indians living with diabetes, plus 136 M with prediabetes

ICMR-INDIAB · Lancet D&E · 2023

62%

of calories in Indian diets come from carbohydrates

ICMR-INDIAB · Nature Medicine · 2025

~270

meals between two quarterly HbA1c tests, none of them seen by the doctor

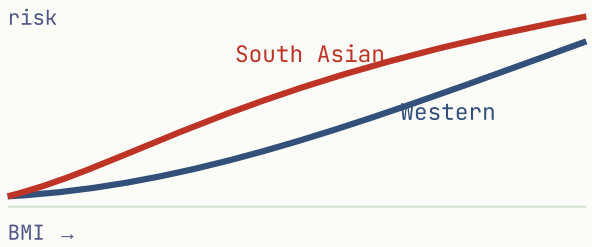
3 meals × 90 days

Post-meal spikes are where the damage happens, and they are invisible to a quarterly lab test. Care stays reactive.

Same diagnosis, different body, different plate.

PHENOTYPE

Insulin resistance at lower BMI



Our cohort: mean BMI 26.6, yet mean fasting glucose 150 mg/dL

PLATE

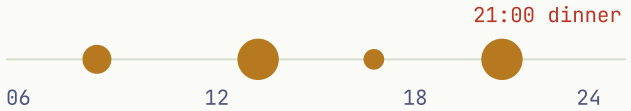
Rice, roti, dosa, chai, sweets

Masala dosa	68 g · GI 77
Rajma chawal	82 g · GI 68
Chicken biryani	90 g · GI 72
Ragi mudde + saaru	60 g · GI 55
Gulab jamun (2)	40 g · GI 80

38 regional dishes modelled with carbs and glycaemic index

ROUTINE

Late dinners, short sleep, festivals



Metformin missed on some days; one bad night lowers next-day insulin sensitivity

OUR ANSWER

One virtual patient, built from two data streams.

Static · EHR

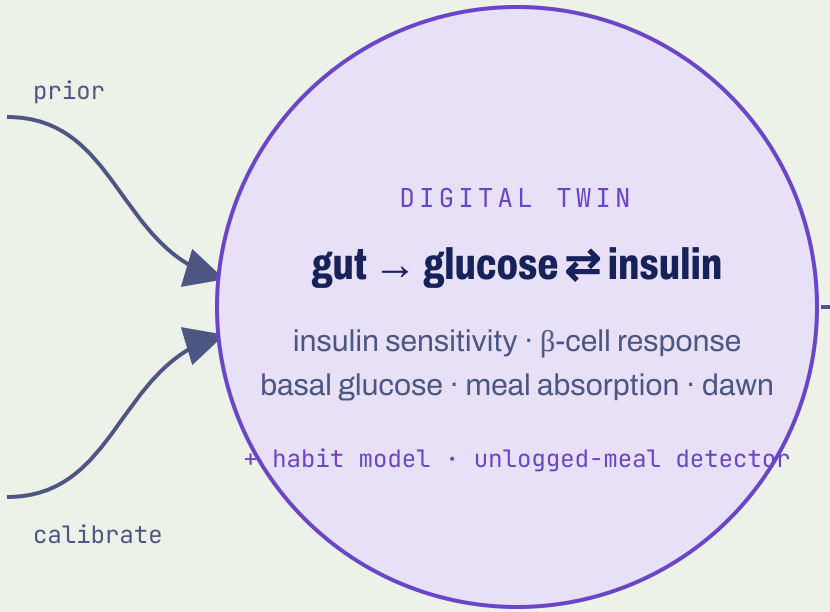
HbA1c, labs, diagnoses, drugs, BMI, TCF7L2 genotype

FHIR R4 · LOINC · SNOMED

Dynamic · wearables

CGM, steps, heart rate, HRV, sleep stages, meals, doses

every 5 minutes



Predict

P(spike > 180 in 2 h) every 15 min

Explain

meal, routine, sleep, activity, dose

Simulate

what if: meal swap, walk, sleep, metformin

DATA · SANDBOX RULES

A synthetic Indian cohort, built from physiology and made realistically messy.

150

patients across 13 cities
114 T2D · 36 prediabetes

604,800

five-minute wearable readings
over 14 days

0

real patients
DPDP Act and HIPAA safe by design

`CGM lag, noise, dropouts` `45–98% of meals logged` `carbs under-estimated` `missed metformin` `short sleep lowers insulin sensitivity` `festival day` `FHIR R4 export`

CGM is simulated from each patient's hidden physiology; HbA1c and fasting glucose are derived back from it, so both streams agree. The hidden truth is kept aside to check the twin.

CHAYA *छाया* · system architecture

Two data streams in, one calibrated virtual patient, 2-hour spike prediction and what-if simulation out

1 · DATA STREAMS

Static / historical: EHR

FHIR R4 Bundle per patient

- Demographics, city, diet
- Diagnoses (SNOMED): T2D, HTN, NAFLD
- Labs (LOINC): HbA1c, FPG, lipids, eGFR
- Medications: metformin, glimepiride, DPP-4i, SGLT2i, insulin
- Genetics: TCF7L2 rs7903146

Dynamic / real-time: wearables

- CGM glucose, every 5 min
- Steps, heart rate
- Nightly HRV (RMSSD), sleep stages
- Meal log (Indian dishes, carbs, GI)
- Medication log

Synthetic cohort

150 Indian patients × 14 days · 604,800 sensor rows · physiology-driven generator. No real patient data.

2 · INGEST & HARMONISE

FHIR parser

LOINC / SNOMED codes → patient profile vector

Stream alignment

5-min grid · sensor gaps kept as gaps · meal and dose events placed on the timeline

Model inputs

Minute-level carbs, GI-dependent absorption, steps, sleep-debt and missed-dose modifiers

3 · DIGITAL TWIN CORE

Per-patient physiological model

`gut → plasma glucose ⇄ insulin action`
`extended Bergman minimal model`

1. **EHR prior.** BMI, waist, HbA1c, duration, TCF7L2, drugs → starting physiology
2. **Calibrate.** MAP fit to 7 days of CGM: insulin sensitivity, β-cell response, basal glucose, meal bioavailability, dawn effect
3. **Observe.** Runs beside live CGM; corrects its state and detects unlogged meals
4. **Habit model.** Learns when and how much this patient usually eats
5. **Forecast & what-if.** Simulates the next 2–4 h under routine or a doctor's scenario

↻ re-calibrates as new data arrives

4 · FUSION & PREDICTION

Fused feature vector

every 15 min

`18 EHR` `25 wearable` `11 twin`

Spike classifier

LightGBM · P(glucose > 180 mg/dL within 2 h)

Glucose regressor

LightGBM · glucose value at +2 h

Explanations

Tree SHAP grouped into clinical drivers: meal, routine, sleep, activity, medication, physiology

5 · DELIVERY

FastAPI on Railway

`PostgreSQL · /patients /simulate · /notes · /pipeline`

Chaya web app

- Home with live forecast slip
- Register, case sheet, stamps
- Reasons, twin physiology
- Rx pad what-if + case notes
- Product tour, case walkthrough

Patient nudge (next)

"Walk 15 min after lunch today"

Evaluation protocol

Patient-level split 105 train / 45 held-out. Temporal split: days 1–7 calibrate, days 8–14 predict. Threshold chosen on training folds only.

Privacy by design

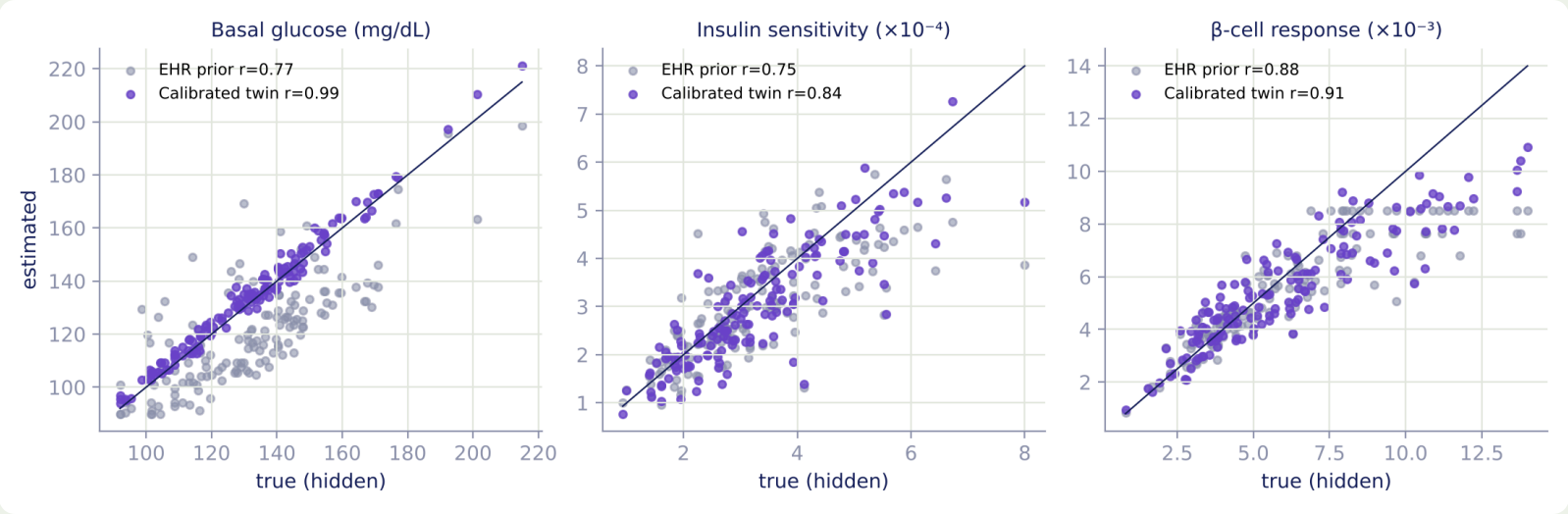
Synthetic data only (DPDP Act 2023 / HIPAA sandbox rules). FHIR-native so it can plug into ABDM-consented records later.

Stack

Python · NumPy/Numba · SciPy · LightGBM · FastAPI · SQLAlchemy · PostgreSQL on Railway · vanilla JS + D3

DOES THE TWIN LEARN THE PERSON?

A week of wearable data recovers each patient's hidden physiology.



BASAL GLUCOSE

0.99 vs 0.77 EHR

INSULIN SENSITIVITY

0.84 vs 0.75 EHR

B-CELL RESPONSE

0.91 vs 0.88 EHR

Correlation with true hidden parameters · 150 patients

RESULTS · 45 HELD-OUT PATIENTS

Eight in ten spikes, flagged before they happen.

0.942
AUROC, spike > 180 mg/dL within 2 hours

83%
sensitivity at 79% precision

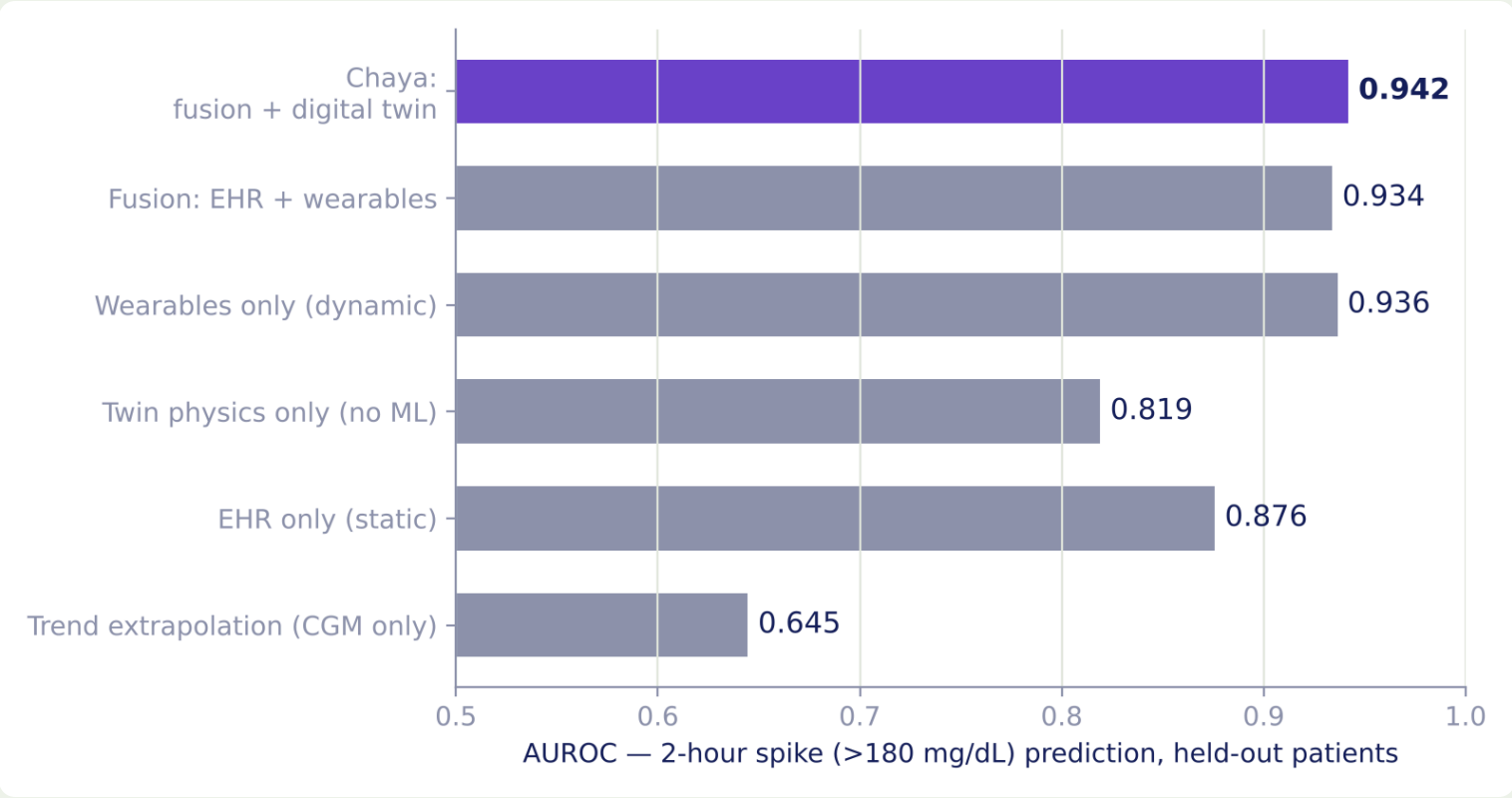
87%
of spikes flagged ≥ 60 min ahead

24
mg/dL error on 2-h glucose (53 if glucose is assumed flat)

Patient-level split: 105 train / 45 test. Days 1–7 calibrate, days 8–14 predict. Threshold set on training folds only. Clarke error grid A+B: 99.3%.

FUSION MATTERS

Each stream adds signal; the twin adds physiology.



Twin physics, zero training: 0.819, far above the CGM trend at 0.645. It works from day one.

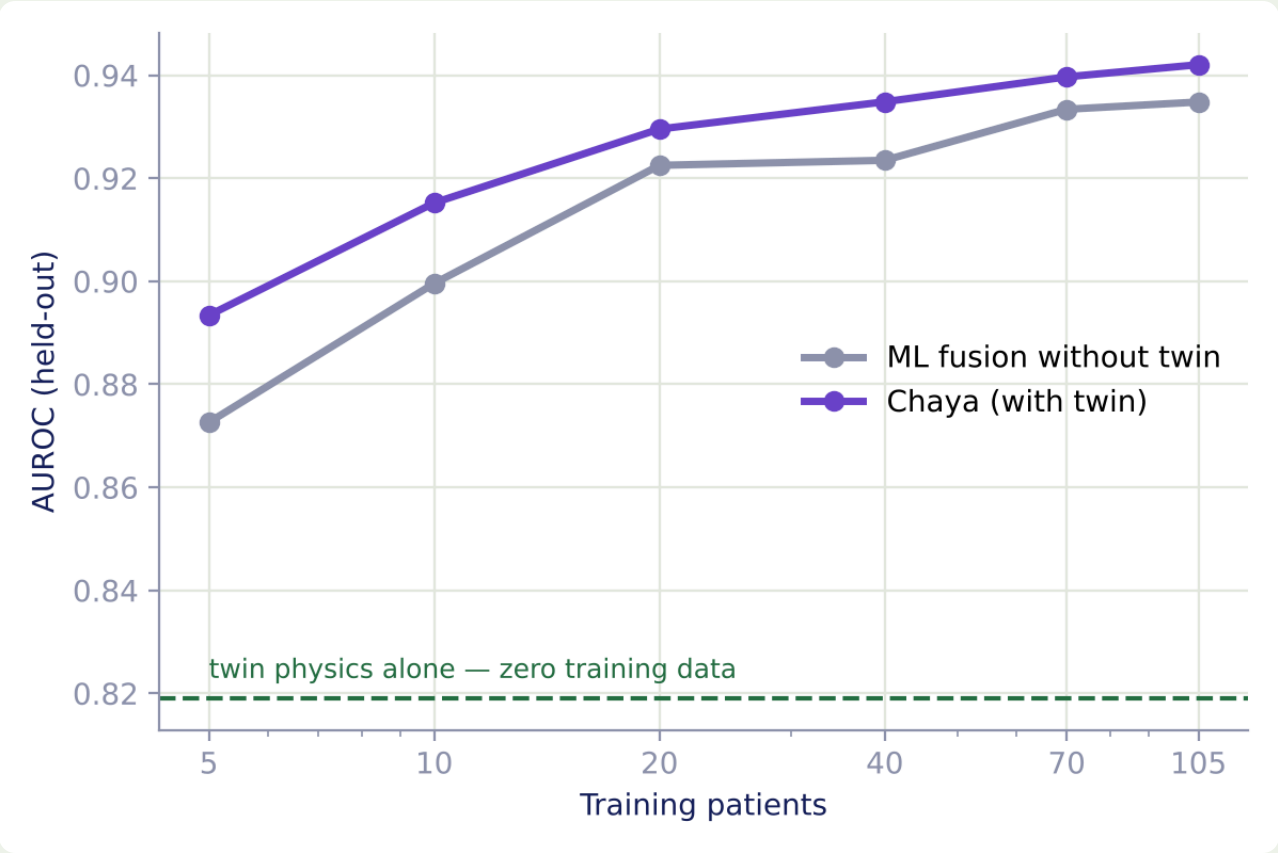
EHR alone knows who is at risk; **wearables** know when.

Fusion + twin gives the best of both: 0.942.

AUROC on held-out patients

WHEN THE TWIN MATTERS MOST

Physics fills the gap when data is scarce.



With only **5 training patients**, the twin lifts AUROC from `0.873` to `0.893`.

That is the situation of a **new clinic**, a **new patient** or a **rare situation** such as a festival day.

With lots of data, ML alone catches up. The twin's lasting value is what-if simulation, which a black box cannot do.

CHAYA छाया | Metabolic forecasts for Type 2 Diabetes | Home | Clinic | ● LIVE · SQLITE | Guide

SIMULATED DEMO · EVERY PATIENT IS SYNTHETIC · DECISION SUPPORT, NOT FOR CLINICAL USE

Mon 14 | Tue 15 | Wed 16 | Thu 17 | Fri 18 | Sat 19 | **Sun 20** — NOW — 12:45 — ▶ Replay — Replaying synthetic week · Mon 14 to Sun 20 Sep 2026

Register

45 of 45

Search name, city or ID

Most spike alerts this week

Pooja	57 warnings	43F · Jaipur · T2D · A1c 8.0%
Srinivas	55 warnings	60M · Hyderabad · T2D · A1c 8.7%
Fatima	52 warnings	52F · Bhubaneswar · T2D · A1c 8.3%
Suresh	52 warnings	41M · Pune · T2D · A1c 8.1%
Rajesh	47 warnings	65M · Kolkata · T2D · A1c 8.0%
Deepak	46 warnings	42M · Kochi · T2D · A1c 7.9%
Karthik	42 warnings	63M · Pune · T2D · A1c 7.6%

GT-1102 · synthetic

Sanjay

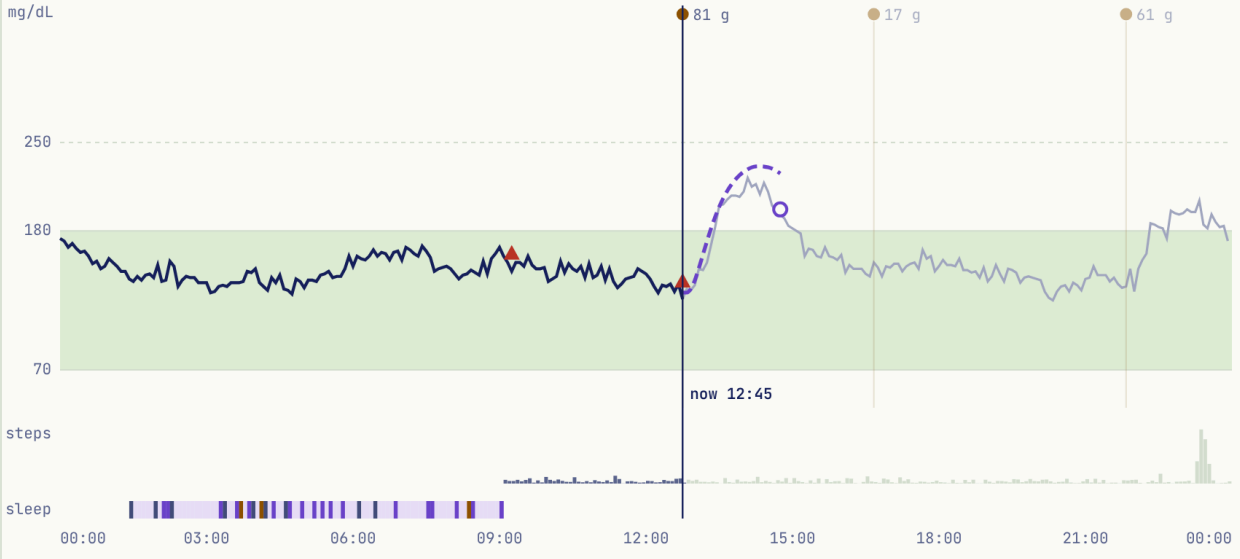
53 · Male · Chennai · Non-vegetarian · BMI 23.9

Type 2 diabetes · 3.9 y | Fatty liver | Metformin | Glimepiride

TIME IN RANGE	ABOVE 180	GMI	VARIABILITY	SPIKE WARNINGS
86 %	14 %	6.9 %	18 % CV	26
meets goal of 70%	above 250: 0%	lab HbA1c 6.4%	stable (goal ≤ 36%)	this week, 2 h ahead

Day trace

Sunday 20 Sep



Next 2 hours

Sun 12:45

SPIKE LIKELY · 92% · NEXT 2 H

92% chance glucose goes above 180 mg/dL by **14:45**.

GLUCOSE IN 2 H	FORECAST PEAK
197 mg/dL	231 at 14:25

WHY CHAYA SAYS THIS

RECENT MEAL
81 g carbs logged 0 min ago (GI 75)

TWIN FORECAST
Twin projects a peak of 231 mg/dL

ACTIVITY
251 steps in the last hour

Mark warning as reviewed

Patient's twin

fits within ±19.9 mg/dL

Insulin sensitivity — 2.19 ×10⁻⁴

EXPLAINABLE BY DESIGN

Every alert comes with its reasons.

Next 2 hours Sun 12:45

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92% · NEXT 2 H

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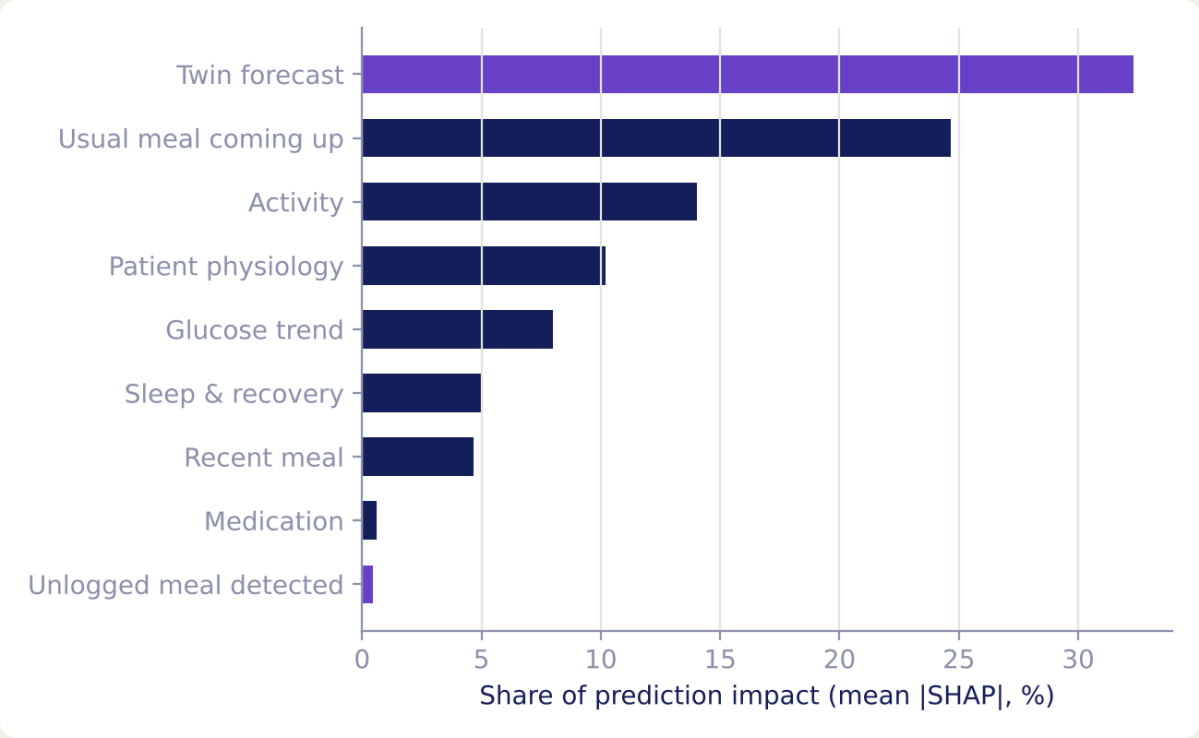
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Tree SHAP contributions grouped into the drivers a clinician thinks in. Violet: drivers that come from the twin.

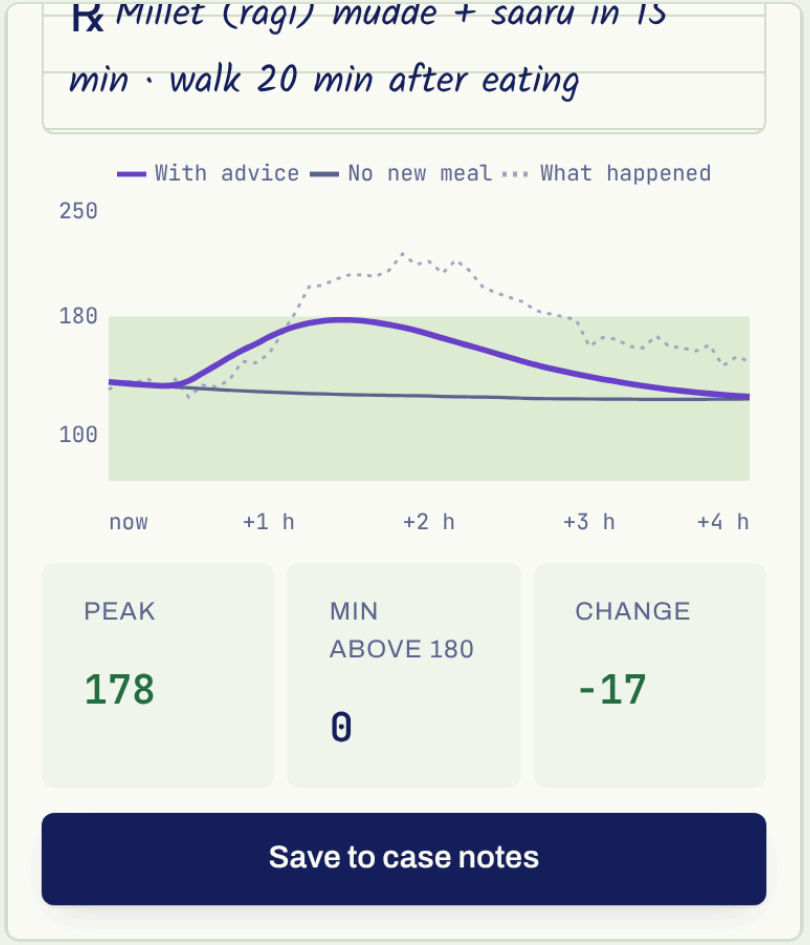
WHAT-IF SIMULATION

Advice in numbers, for this patient's body.

SANJAY · 53 · T2D 3.9 Y · SUNDAY 12:15, BEFORE LUNCH

LUNCH IN 15 MIN	PEAK MG/DL	TIME > 180
Chicken biryani 90 g · GI 72	237	115 min
Biryani + 20-min walk	217	85 min
Ragi mudde + saaru 60 g · GI 55	194	70 min
Ragi mudde + 20-min walk	178	0 min

Same patient, same morning. Swapping the grain and walking after lunch keeps him in range.



Peak glucose over the next 4 h, simulated by the calibrated twin

SAFETY AND PRIVACY

Built to support the doctor, not to replace them.

Clinician in the loop

Decision support only. No dosing. Every alert shows its reasons and the twin's projection.

Alert fatigue managed

Consecutive alerts merge into one episode: about 4 per patient-day, with a threshold the clinic can tune.

DPDP-ready

Synthetic data only. FHIR-native, so it can use ABDM-consented records with purpose limitation.

NEXT

From prototype to clinic.

Now

Working twin, models, API and console

Validate

Public CGM + wearable datasets (CGMacros, BIG IDEAs Lab)

Pilot

Supervised pilot with a partner diabetes clinic

Extend

Patient nudges, hypo risk for sulfonylurea and insulin, hypertension twin

PROACTIVE, NOT REACTIVE

See the spike before it happens. Show the patient how to avoid it.

github.com/r-e-sai-harsha/GlucoTwin